

1. Administrative & Study Identification

Official Study Title: A Phase 1/2, Open-Label, Multicenter Study to Evaluate a Dosing Regimen with Two Step-up Priming Doses and Longer Dosing Intervals of Elranatamab (PF-06863135) Monotherapy in Participants with Relapsed/Refractory Multiple Myeloma **Public Study Title:** A Study to Learn About the Study Medicine (Elranatamab) in Participants With Multiple Myeloma That Has Come Back After Responding to Treatment or Has Not Responded to Treatment **Short Title/Acronym:** MagnetisMM-9 **Protocol Number:** PF-06863135 **NCT Number:** NCT05014412 **Posting ID:** 411866265936246350 **Sponsor / Company Name:** Pfizer **Investigational Product:** Elranatamab (PF-06863135) **Trade Name:** Elrexfio **Phase of Development:** Phase 2 (PHASE2) **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of an alternative step-up dosing approach by determining the Number of Participants With Grade 2 or Higher Cytokine Release Syndrome (CRS) during Cycle 1 (Parts 1 and 2). **Secondary Objectives:** To evaluate dose-limiting toxicities (Part 2A), Number of Participants With Adverse Events (AEs) and Serious AEs (SAEs), and Number of Participants With Maximum Grade 3 or 4 and Grade 5 AEs. Additional secondary assessments include Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), Minimal Residual Disease (MRD) negativity rate, and pharmacokinetics (PK).

2.2 Background & Rationale Elranatamab is a humanized bispecific antibody targeting B cell maturation antigen (BCMA) and CD3, resulting in targeted T cell-mediated cytotoxicity. Cytokine release syndrome (CRS) is a common adverse event associated with bispecific antibodies. This study (MagnetisMM-9) investigates an alternative priming regimen (4 mg and 20 mg step-up doses) and longer dosing intervals to mitigate the incidence and severity of CRS while maintaining clinical efficacy and durable responses in patients with relapsed/refractory multiple myeloma (RRMM).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non-randomized) **Blinding:** Open-label **Randomization:** N/A **Current Status:** Active but not currently recruiting

3.2 Planned Enrollment & Duration Target **N**: 86 patients **Number of Sites**: Multicenter (including centers in the USA, UK, Japan, and Taiwan) **Estimated Study Start**: October 2021 **Estimated Primary Completion**: April 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with a diagnosis of multiple myeloma (IMWG criteria, Rajkumar et al, 2014). 2. Measurable disease, as defined by at least 1 of the following: Serum M-protein > 0.5 g/dL by SPEP; Urinary M-protein excretion > 200 mg/24 hours by UPEP; or Serum immunoglobulin FLC ≥ 10 mg/dL (≥ 100 mg/L) AND abnormal serum immunoglobulin kappa to lambda FLC ratio. 3. Refractory to at least one proteasome inhibitor (PI), one immunomodulatory drug (IMiD), and one anti-CD38 antibody. 4. Relapsed/refractory to the last anti-myeloma regimen with an ECOG performance status ≤ 1 . 5. Resolved acute effects of any prior therapy to baseline severity or CTCAE Grade ≤ 1 . 6. Not pregnant and willing to use contraception.

4.2 Exclusion Criteria 1. Smoldering multiple myeloma, active plasma cell leukemia, POEMS syndrome, amyloidosis, or Waldenström's macroglobulinemia. 2. Known active CNS involvement or clinical signs of myelomatous meningeal involvement. 3. Stem cell transplant within 12 weeks prior to enrollment or active GVHD. 4. Active HBV, HCV, SARS-CoV2, HIV, or any active, uncontrolled bacterial, fungal, or viral infection. 5. Any other active malignancy within 3 years prior to enrollment, except for adequately treated basal cell or squamous cell skin cancer, or carcinoma in situ or Stage 0/1 malignancy with minimal risk of recurrence per investigator. 6. Previous treatment with an anti-BCMA bispecific antibody or CAR-T cell therapy. 7. Live attenuated vaccine within 4 weeks of the first dose. 8. Previous administration with an investigational drug within 30 days or 5 half-lives preceding the first dose of study intervention used in this study (whichever is longer). 9. Known or suspected hypersensitivity to the study intervention, or any of its excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: In Cycle 1, patients receive premedications and two step-up priming doses on Day 1 (4 mg) and Day 4 (20 mg). Following priming, patients receive 76 mg once weekly (QW) for the next 6 cycles (Part 1). Subsequent cycles explore less frequent dosing intervals (Q2W or Q4W) and higher doses (e.g., 116 mg or 152 mg in Part 2). **Route of Administration**: Subcutaneous (SC) injection. **Duration of Treatment**: Continued until disease progression or unacceptable toxicity. Participation will be at least two years.

5.2 Concomitant & Prohibited Therapy Permitted: Premedications to mitigate Cytokine Release Syndrome (CRS) before the initial step-up doses. **Prohibited:** Other systemic anti-myeloma therapies and live vaccines during the trial period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Disease Assessment (SPEP/UPEP/FLC)
Baseline / Cycle 1	Days 1, 4, 8	Step-up priming doses (4 mg, 20 mg), First full dose (76 mg), Strict CRS safety monitoring
Interim	QW, Q2W, or Q4W	Ongoing SC injections, Efficacy Assessments (IMWG response), PK/PD sampling, AE review
End of Study	Until Progression	Final vital signs, safety follow-up, evaluation of PFS and OS

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of Participants With Grade 2 or Higher Cytokine Release Syndrome (CRS) During Cycle 1 (Parts 1 and 2). **Measurement Method:** Clinical assessment and laboratory evaluation utilizing ASTCT criteria (Graded 1 to 5 based on fever, hypotension, and hypoxia). Organ toxicities associated with CRS graded according to CTCAE v5.0.

7.2 Secondary & Exploratory Endpoints * Number of Participants With Dose Limiting Toxicities (DLTs): Part 2A. * Number of Participants With Adverse Events (AEs) and Serious AEs (SAEs) [All Causalities and Treatment Related]: Parts 1 and 2. * Number of Participants With Maximum Grade 3 or 4 and Grade 5 AEs [All Causalities and Treatment Related] Per National Cancer Institute (NCI) CTCAE v5: Parts 1 and 2. * Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS). * Minimal Residual Disease (MRD) negativity rate and Pharmacokinetics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Focus on CRS (incidence and severity via ASTCT criteria), neurologic toxicities (ICANS), and infection tracking after each SC dose. **Serious Adverse Events (SAEs):** Routine safety reporting protocol, with heightened 24-hour observation periods surrounding the initial priming doses. **Data Monitoring Committee (DMC):** Internal/External safety review to determine dose escalations and cohort expansions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy evaluations mapped to the treated population; Safety Set mapped to any patient receiving at least one dose of elranatamab. **Sample Size Justification:** Estimated initial target powered to detect differences in CRS rates versus prior registrational studies. **Handling of Missing Data:** Standard survival methodology, employing censoring for PFS and OS endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite monitoring during the Cycle 1 step-up dose phase; regular assessments throughout Q2W and Q4W maintenance phases. **Audit Trail:** Validated eCRF locking and independent clinical data clarification reviews.

11. Study Identifiers & Governance

Primary ID: {{PF-06863135}} **Posting ID:** {{411866265936246350}}
Registry Number: {{NCT05014412}} **Sponsor/Lead Organization:** {{Pfizer}} **Official Regulatory Title:** {{A Phase 1/2, Open-Label, Multicenter Study to Evaluate a Dosing Regimen with Two Step-up Priming Doses and Longer Dosing Intervals of Elranatamab (PF-06863135) Monotherapy in Participants with Relapsed/Refractory Multiple Myeloma}} **Public Study Title:** {{A Study to Learn About the Study Medicine (Elranatamab) in Participants With Multiple Myeloma That Has Come Back After Responding to Treatment or Has Not Responded to Treatment}}

12. Clinical Framework & Ontology

Primary Condition / Disease Area: {{Multiple Myeloma}} **Preferred UMLS Name:** {{Relapsed Refractory Multiple Myeloma}} **Semantic Type:** {{Neoplastic Process}} **Disease Definition:** {{A bone marrow-based plasma cell neoplasm characterized by a serum monoclonal protein and skeletal destruction with osteolytic lesions, pathological fractures, bone pain, hypercalcemia, and anemia. Clinical variants include non-secretory myeloma, smoldering myeloma, indolent myeloma, and plasma cell leukemia. (WHO, 2001)}} **Ontology Codes:** - **MeSH Code:** D009101 - **HPO Code:** HP: 0006775 - **SNOMED ID:** 109989006 (Hierarchy: 269475001, 415111003, 1162576007, 64572001) **Medical Methodology:** {{BCMA-CD3 Bispecific Antibody Therapy}} **Drug Trade Name:** {{Elrexfio}} **EMA Product Information:** {{https://www.ema.europa.eu/en/documents/product-information/elrexfio-epar-product-information_en.pdf}}

13. Study Procedures & Methodology

Management Pathway: {{RRMM Treatment Pathway}} **Education Protocol (HCP):** {{CRS and ICANS Management Training}} **Education Protocol (Patient):** {{Injection site and AE symptom reporting}} **Quality Audit Mechanism:** {{Patient response tracking and PK/PD monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{At least two years (until disease progression)}} **Onsite Visit Cadence:** {{Cycle 1 (Days 1, 4, 8, 15, 22), then QW, Q2W, or Q4W}} **Remote Monitoring Frequency:** {{As needed between clinic injection visits}} **Checklist Review Interval:** {{Every 28-day cycle}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				